International and Monetary Economics

Course Code: 30054

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Lecture 13

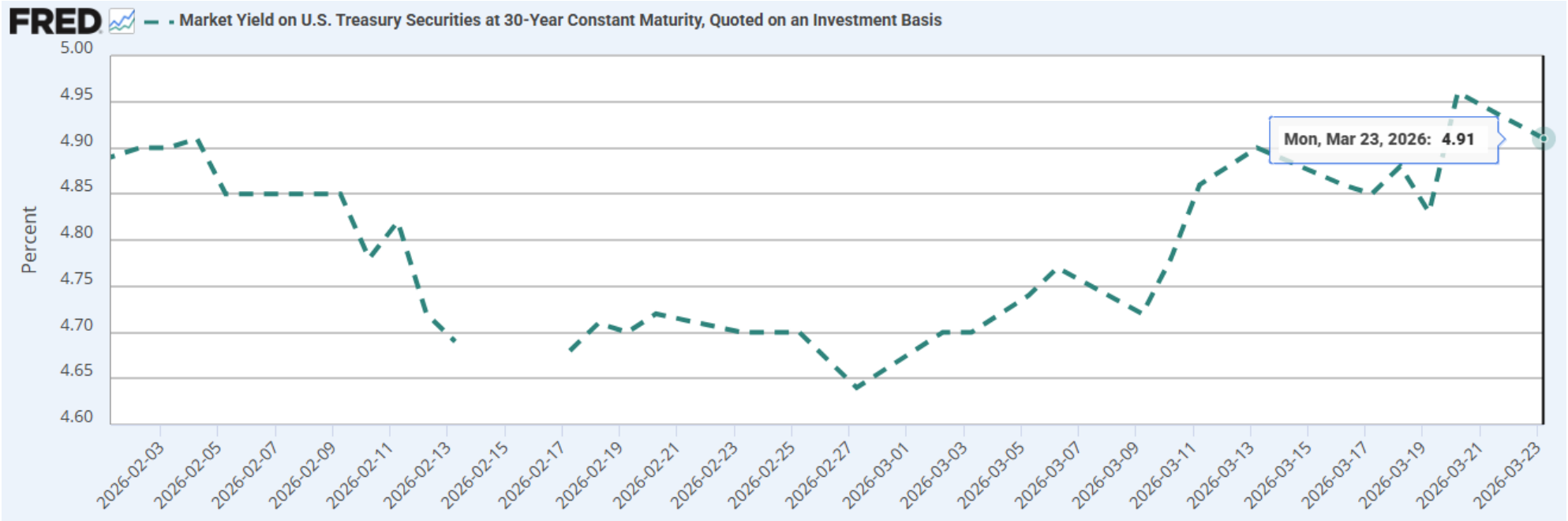
Roadmap

Last time

1. Policy exercises
2. Liquidity trap

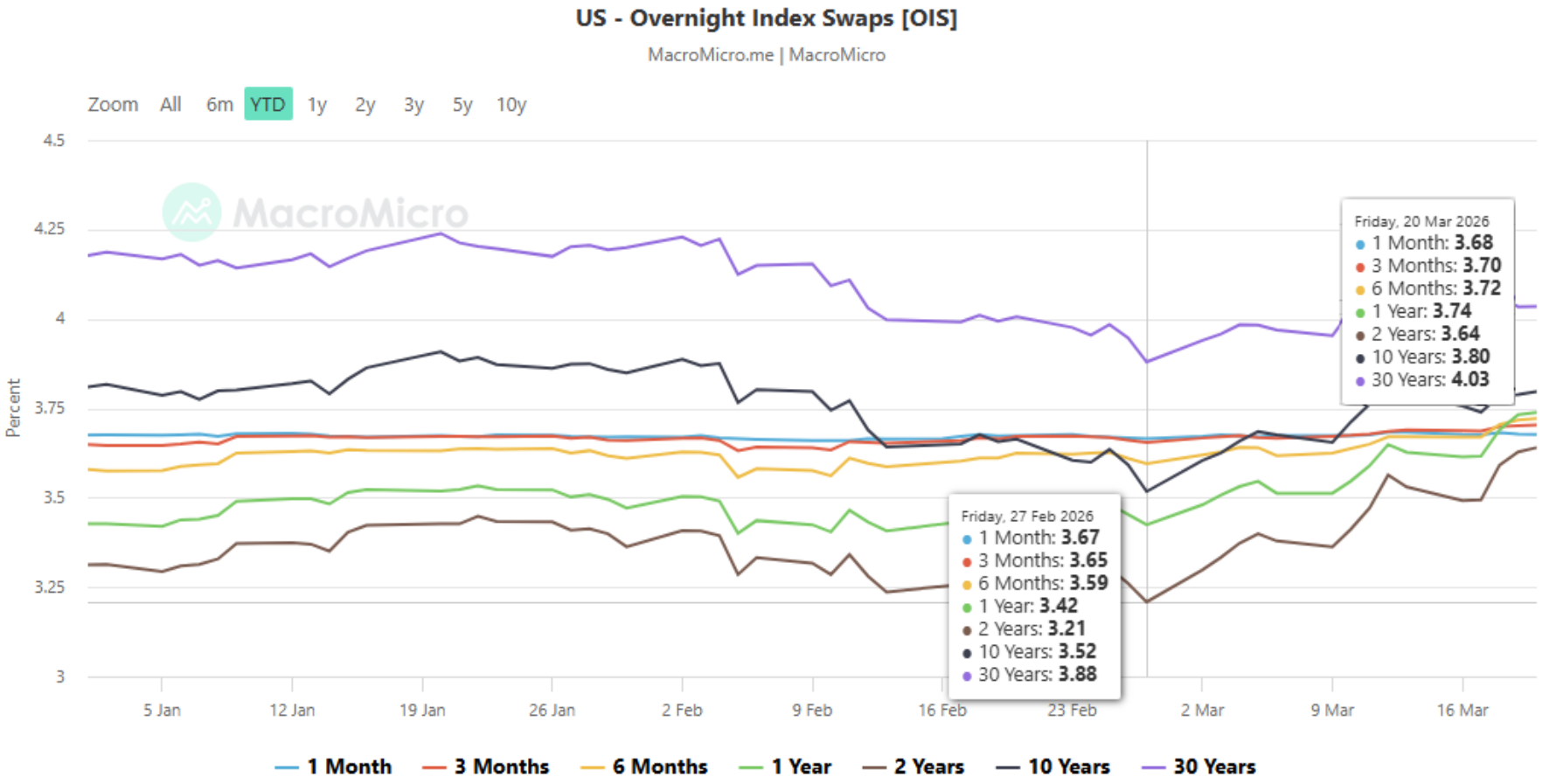
Today

1. The XX curve
2. PCP, LCP, and DCP
3. Fixed exchange rate



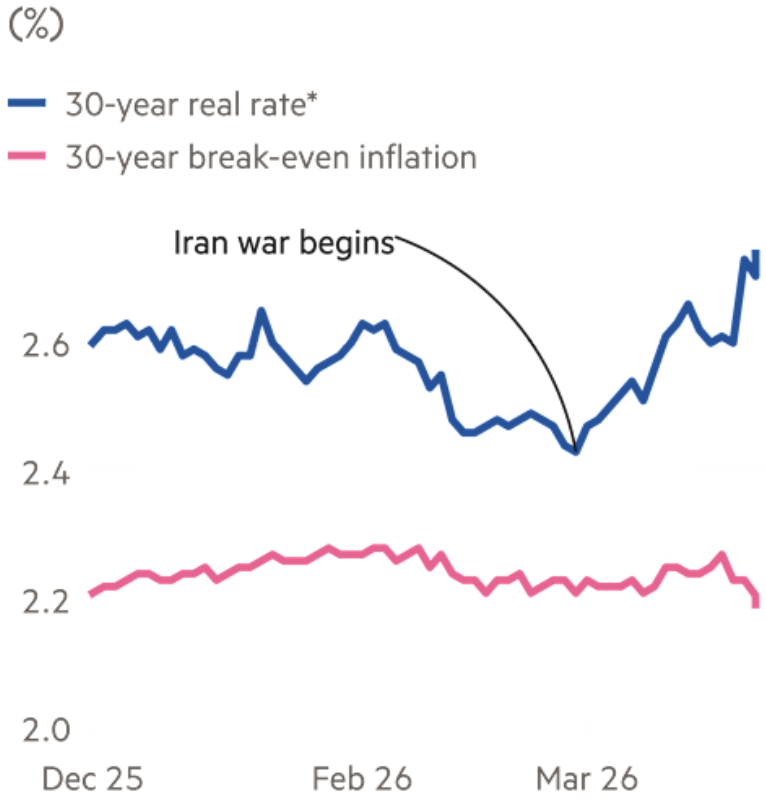
Why have the yields on the U.S. long-term government bonds gone up?

Markets expect higher Federal Funds Rate



Note: An Overnight Index Swap (OIS) reflects market expectations of future short-term nominal interest rates because it exchanges a fixed rate for the realized compounded overnight rate (e.g., SOFR) over the life of the contract; under no-arbitrage, the fixed rate must equal the market's risk-adjusted expectation of that future overnight rate path. Since the floating leg is directly tied to the actual policy-relevant overnight rate, and credit risk is minimal, OIS rates provide a clean measure of the expected path of short-term nominal interest rates (up to a small term premium).

Long-term Inflation Expectations and Real Rates



*as proxied by 30 year inflation-protected Treasury yield
Source: Federal Reserve

*as proxied by the 10 year inflation protected Treasury yield

Why are real rates going up?

A potential answer: **higher risk premium:** a higher chance of future AS shocks that both increase inflation and reduce economic activity)

Source: FT

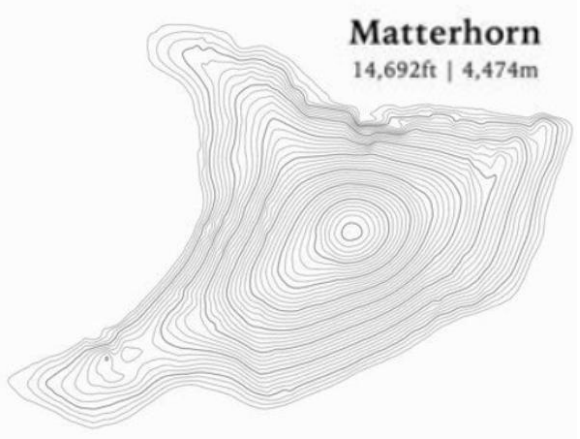
Policy and Trade Balance

How does NX change in our policy experiments?

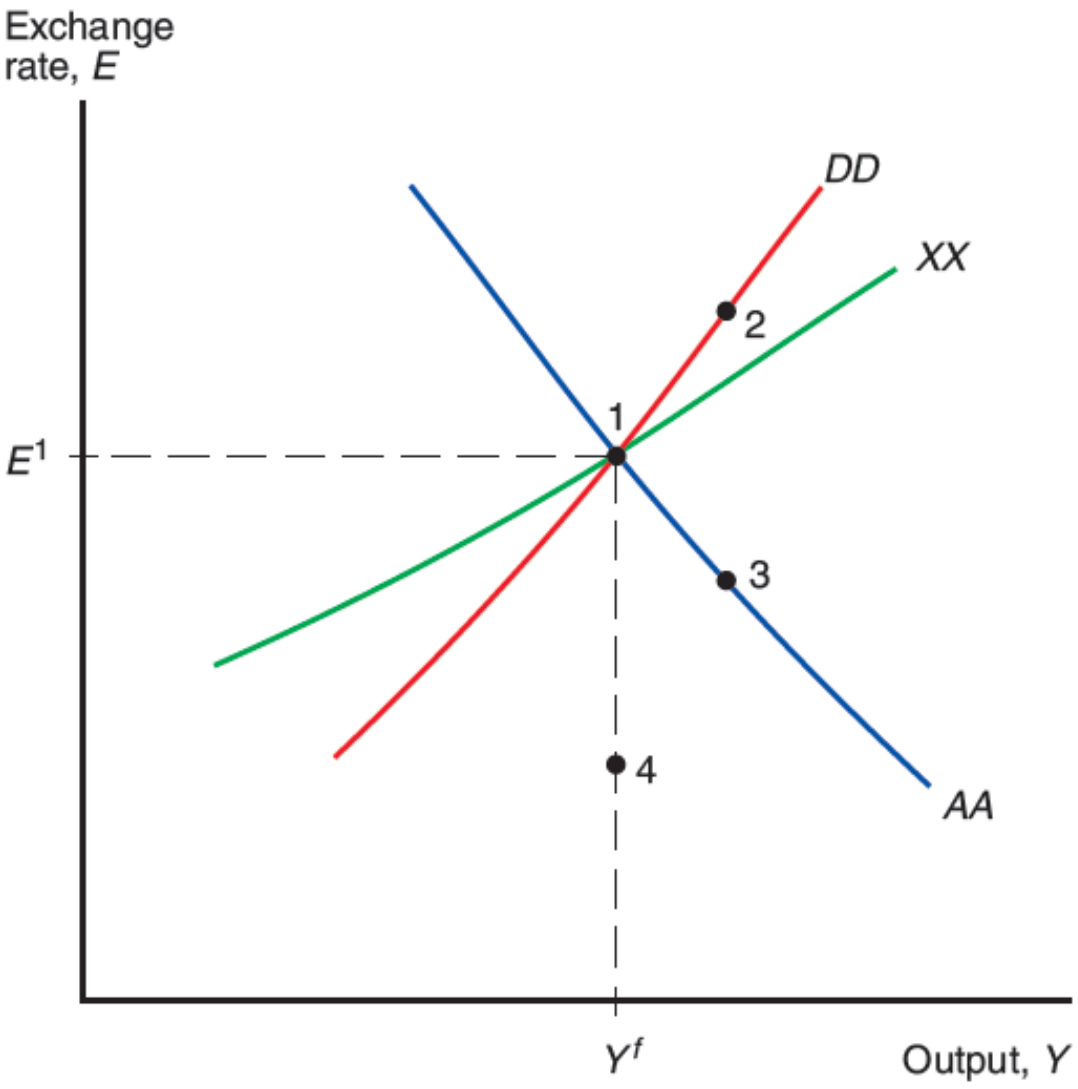
- Policy makers are often concerned about trade balance
  - Negative economic consequences
  - Political pressure
- A useful tool
  - The XX schedule shows combinations of the exchange rate and output at which the NX balance would be equal to some level X

$$NX\left(\frac{EP^*}{P}, Y - T\right) = X$$

  - This is like a topographical map that shows levels of constant elevation

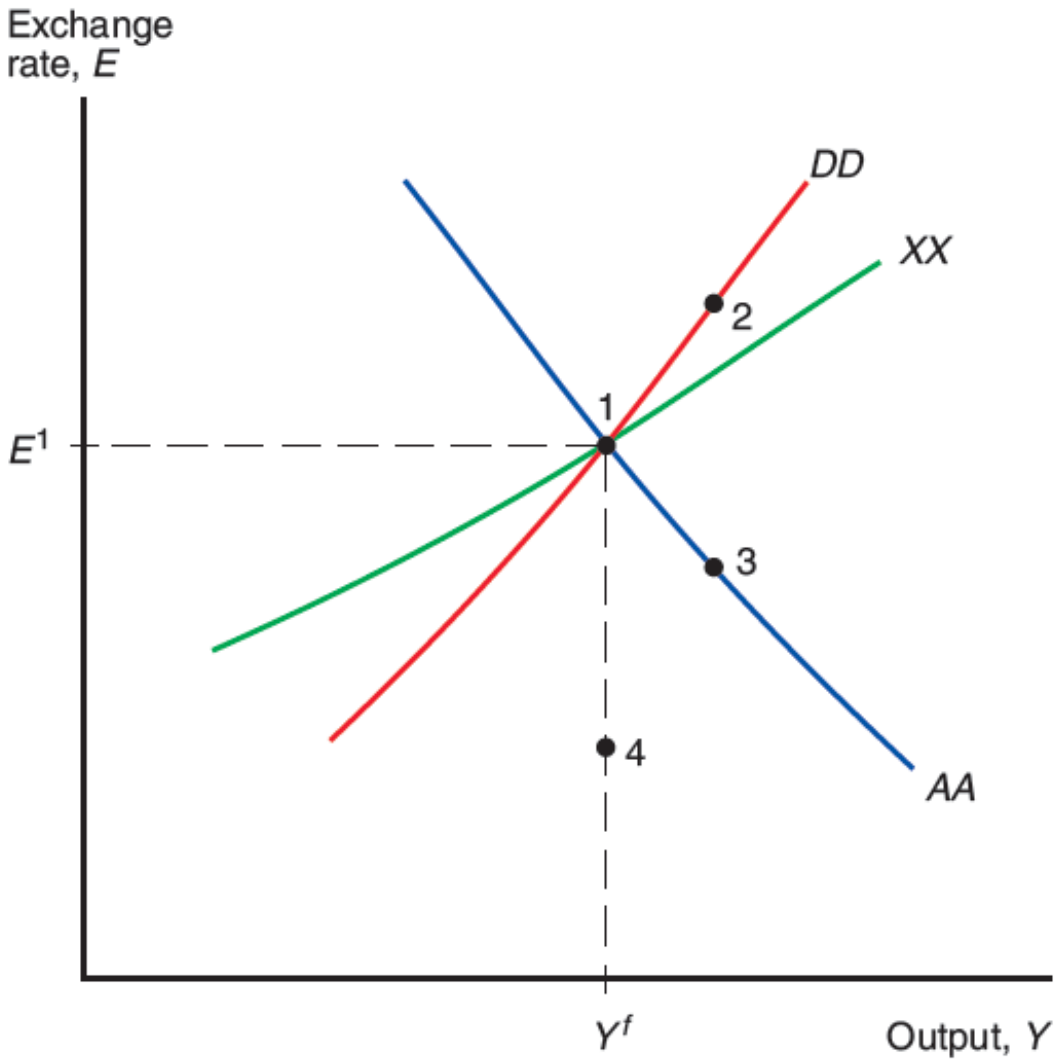


The XX Schedule



- Upward sloping
- Flatter than DD
  - it is easy to understand if you write the equation for the DD curve as Y-C(Y-T)-G-I=NX(EP*/P,Y-T). If Y goes up, then the left-hand side of this equation goes up, and hence, the right-hand side must go up. So, NX increases along the DD curve. As a result, the XX curve is flatter than the DD curve.
- $NX > X$ above the XX schedule, $NX < X$ below the XX schedule

The XX Schedule



- **Monetary expansion**
  - movement from 1 to 2
  - improves NX
- **Fiscal expansion**
  - movement from 1 to 3 (a temporary change)
  - reduces NX

PCP, LCP, and DCP

Exchange Rates and Real Economy

- Classic view
  - ER depreciation boosts demand for domestically produced goods
    - sometimes called the ***expenditure switching effect.***

- Which currency is used to price internationally traded goods?
  - PCP (producer currency pricing)
  - LCP (local currency pricing)
  - DCP (dominant currency paradigm)

Trade Balance and Prices

- Consider trade between the Eurozone (home) and the UK (foreign country)
- Eurozone net exports

$$NX = Ex\left(\frac{P^*}{P^{\text{Im from EA in £}}}\right) - \frac{P^{\text{Im from UK in €}}}{P} Ex^*\left(\frac{P}{P^{\text{Im from UK in €}}}\right)$$

- Note: *this formula does not take into account that*
  - the price of home consumption basket P differs from the price of domestically produced goods (that is, CPI is not equal to PPI);
  - the price of domestic goods sold abroad differs from the price of all domestically produced goods (because not all produced goods are traded internationally).

PCP

Producer-Currency Pricing (PCP)

The assumption used so far

- Traded goods prices are set and **sticky** in producers' currency
- Example
  - Prada store in London adjusts prices to movements in EUR-GBP FX
- Formally

$$P^{\text{Im from EA in £}} = \frac{P}{E}, \qquad P^{\text{Im from UK in €}} = P^{*}E$$

  - **(Bilateral) exchange rate pass-through** (i.e., how much FX changes are transmitted to prices) is 1
  - Trade balance is

$$NX = Ex\left(\frac{P^{*}E}{P}\right) - \frac{P^{*}E}{P}Ex^{*}\left(\frac{P}{P^{*}E}\right)$$

Government Policies in the PCP world

- Domestic (Eurozone) policies
  - A policy that depreciates the domestic currency increases net exports and improves GDP (under our assumption that the volume effect dominates the value effect)
  - A basis for **currency war** concerns that policy-makers often express:
    - A depreciation of domestic currency "steals" demand from a foreign country
- Foreign (U.K.) policies
  - A policy that depreciates foreign currency reduces domestic net exports
- U.S. policies
  - US monetary policy does not have **a direct effect** on net exports from the Eurozone to the UK

LCP

Local-Currency Pricing

- Traded goods prices are set and **sticky** in consumers' currency

- Example
  - Prada store in London keeps its prices set in pounds without any relation to the Euro-pound exchange rate

- Formally

$$NX = Ex\left(\frac{P^*}{P^{\text{Im from EA in £}}}\right) - \frac{P^{\text{Im from UK in €}}}{P} Ex^*\left(\frac{P}{P^{\text{Im from UK in €}}}\right)$$

  - **(Bilateral) exchange rate pass-through** is 0.

Government Policies in the LCP world

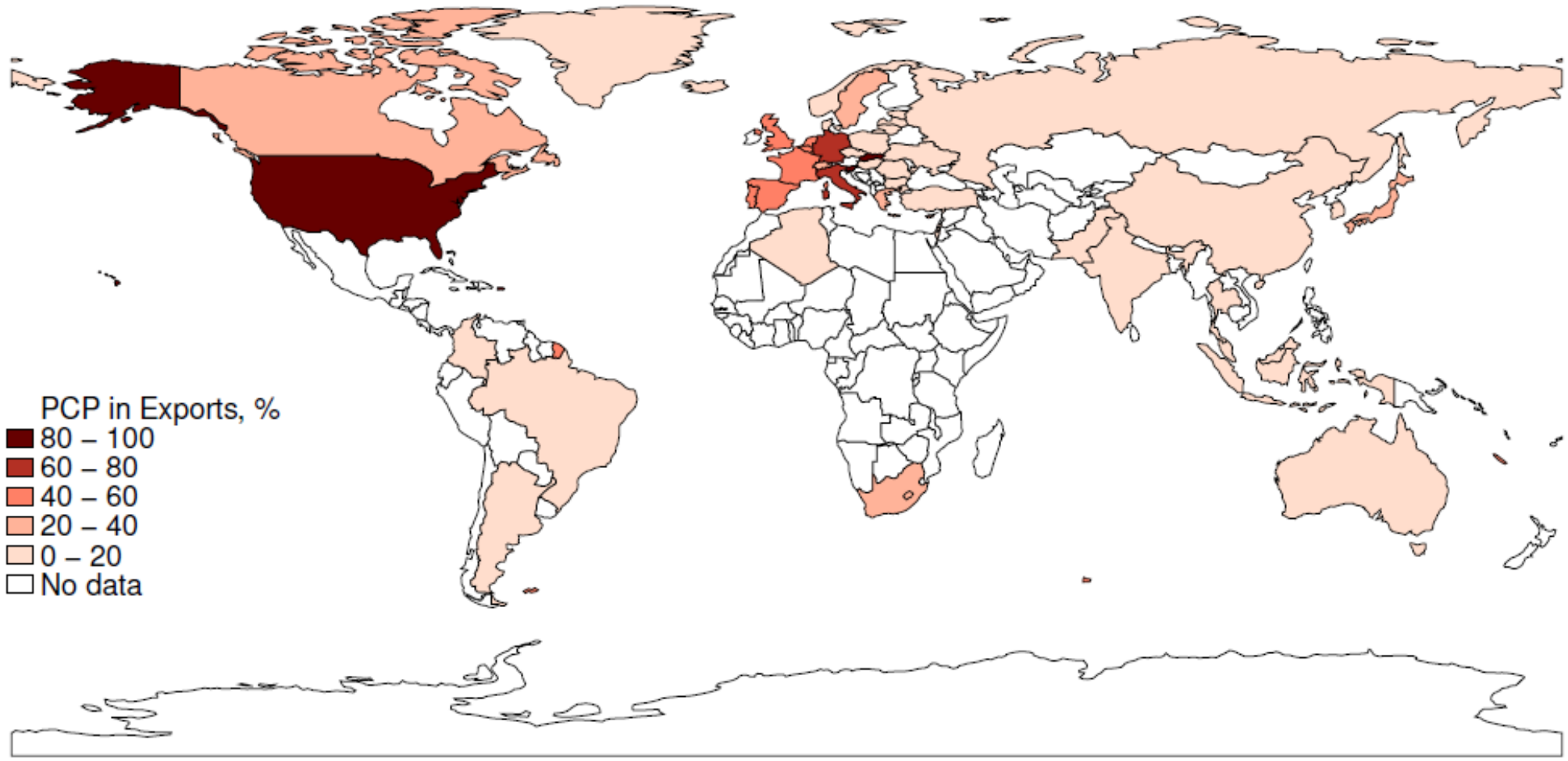
- Domestic (Eurozone) policies
  - A policy that depreciates the domestic currency does not have a direct effect on net exports and GDP
- Foreign (U.K.) policies
  - A policy that depreciates foreign currency does not have a direct effect on domestic net exports
- U.S. policies
  - Monetary policy of a third country does not have a direct effect on net exports from the Eurozone to the UK

DCP

Dominant Currency Paradigm = Dollar Currency Pricing

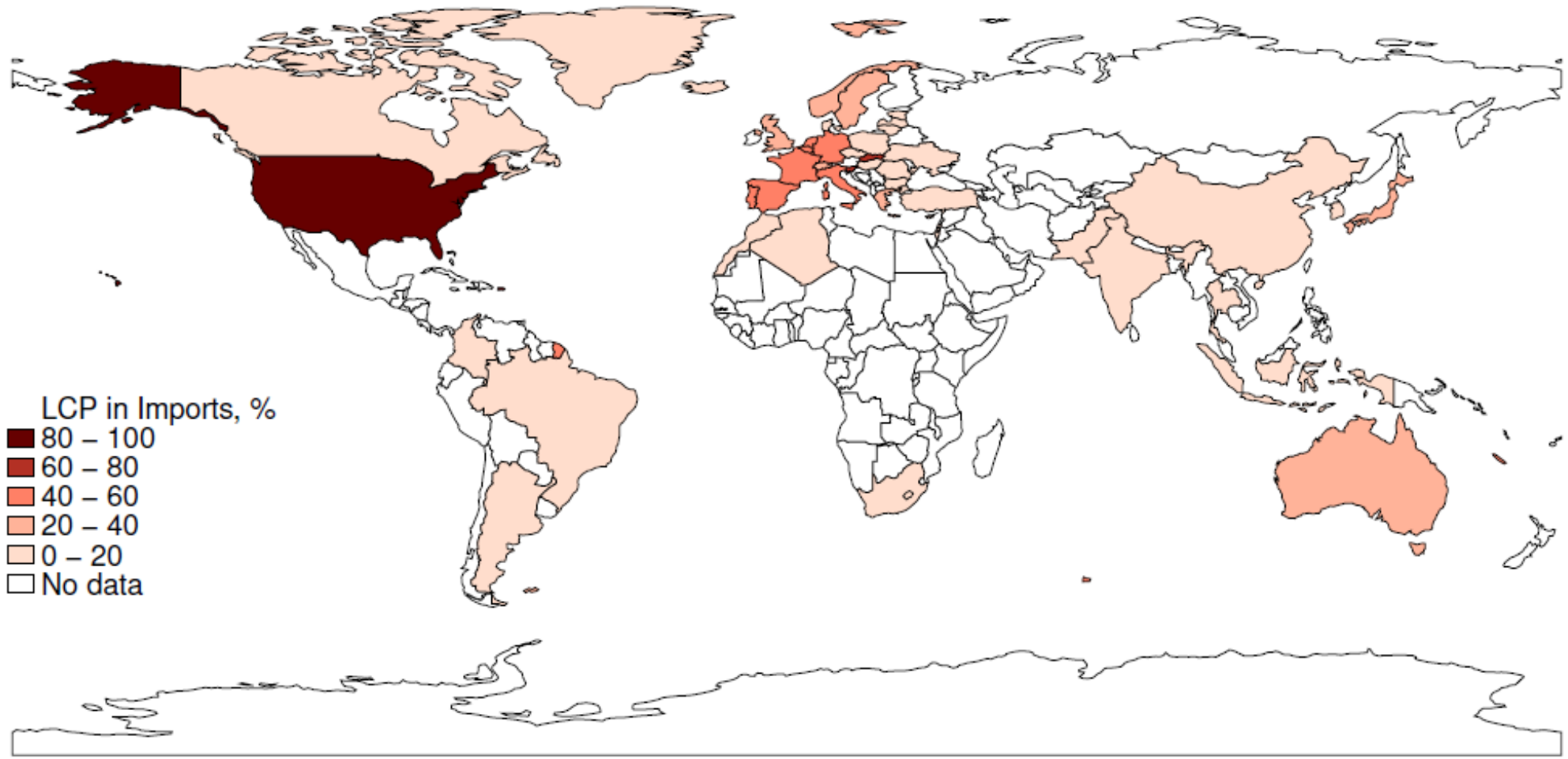
- **Gita Gopinath**: a prof at Harvard and a former First Deputy Managing Director of the IMF
  - 2018 World Bank lecture (on YouTube)
  - Gopinath, “The International Price System” (2015) [PDF on blackboard]

Producer Currency Pricing (PCP) in Exports



Data sources: Gopinath (2016), Kamps (2006), Lai and Yu (2014), Sokolova (2015)

Local Currency Pricing (LCP) in Imports



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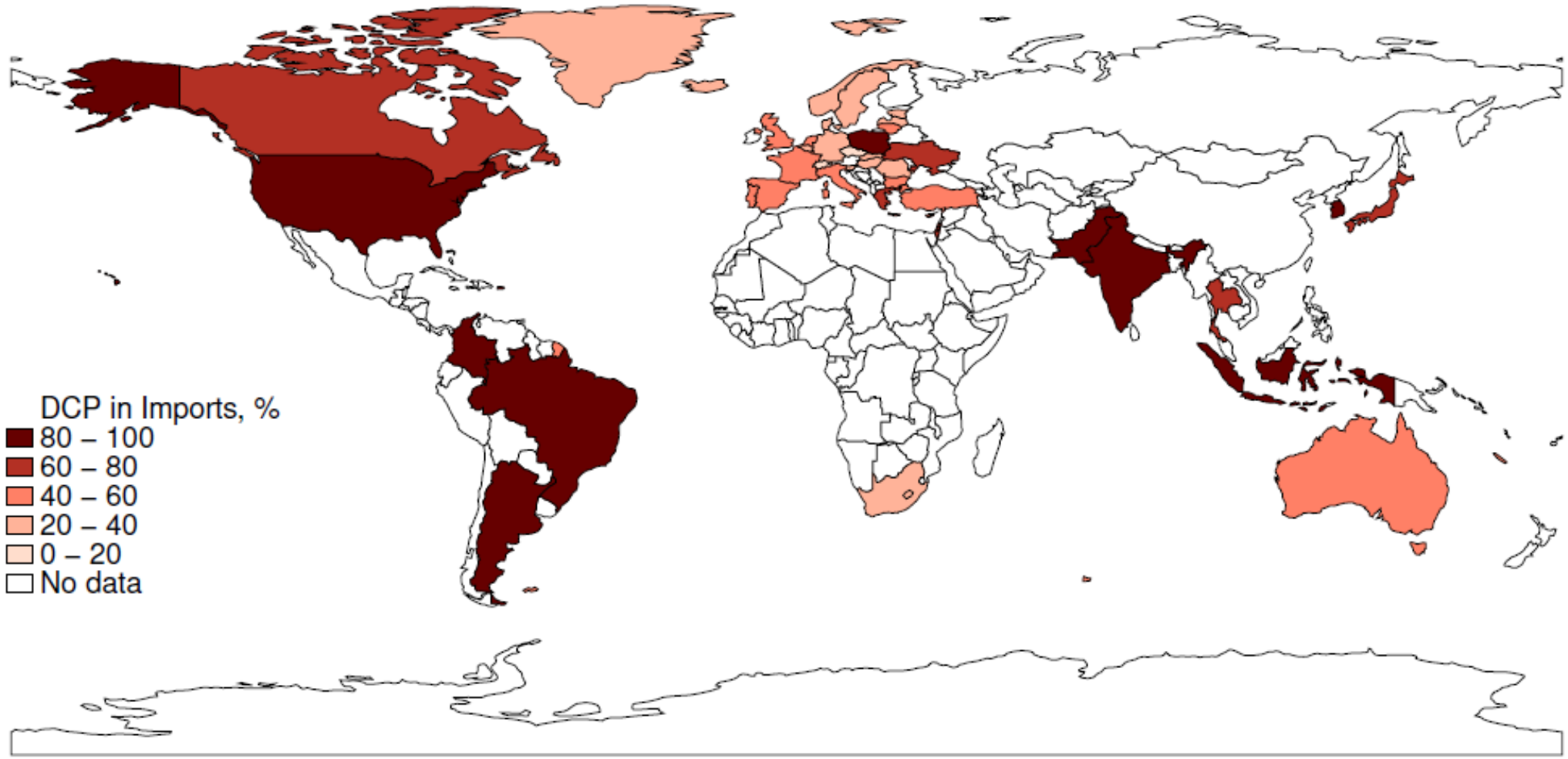
Dollar Currency Pricing (DCP) in Imports



Data sources: Gopinath (2016), Kamps (2006), Lai and Yu (2014), Sokolova (2015)

Dominant Currency Paradigm

- Traded goods prices are **sticky** in a small set of dominant currencies (most notably, the U.S. dollar)
- Example
  - Prada store in London keeps its prices sticky in USD and adjusts its pound price to variations in the USD-GBP exchange rate
- Formally

$$P^{\text{Im from EA in £}} = P^{\text{Im from EA in \$}} \cdot E_{£/\$}$$

$$P^{\text{Im from UK in €}} = P^{\text{Im from UK in \$}} \cdot E_{€/\$}$$

  - **(Bilateral) exchange rate pass-through** is 0.
  - **(Dollar) exchange rate pass-through** is 1.
- And

$$NX = Ex\left(\frac{P^*}{P^{\text{Im from EA in \$}} \cdot E_{£/\$}}\right) - \frac{P^{\text{Im from UK in \$}} \cdot E_{€/\$}}{P} Ex^*\left(\frac{P}{P^{\text{Im from UK in \$}} \cdot E_{€/\$}}\right)$$

Government Policies in the DCP World

- Domestic (Eurozone) policies
  - A policy that depreciates domestic currency does not have a direct effect on exports, but it can influence imports ⇒ the benefits of depreciation are not as big as under PCP
- Foreign (U.K.) policies
  - A policy that depreciates foreign currency does not have a direct effect on domestic imports, but it can affect domestic exports
- U.S. policies
  - The monetary policy of a dominant currency country affects trade between the Eurozone and the UK
  - A dollar appreciation reduces EA exports to the UK and imports from the UK to the EA (if the volume effect dominates the value effect)

What explains DCP?

Potential answers

1. FX risk in supply chains
   - Much of international trade is in intermediate goods. If my firm uses intermediate inputs priced in dollars, then I have incentives to set and keep prices sticky in dollars as well. This reduces my FX risk.
2. Strategic complementarities in price setting
   - If my competitor sets prices in dollars, I have incentives to also set prices in dollars to avoid unwanted fluctuations in my prices relative to my competitors' prices.
3. Finance
   - There is a huge demand for $ safe assets in the world ⇒ the returns on these assets are low ⇒ when I set my prices in $, I can use my invoices as collateral when borrowing cheaply in $

Fixed Exchange Rate

Why Talk about Fixed Exchange Rates?

1. Lessons of the past for the future
   - Many fixed ER historical episodes can shed light on policy proposals
2. Managed floating
   - Most countries smooth short-run FX fluctuations
3. Regional currency arrangements
   - European exchange rate mechanism (1979-99)
   - European union's exchange rate mechanism (Denmark)
4. Developing countries and countries in transition
   - often fix their currencies to $ or €

IMF Monetary Policy Frameworks Classification

- Exchange rate anchor
  - "The monetary authority buys or sells foreign exchange to maintain the exchange rate at its predetermined level or within a range. The exchange rate thus serves as the nominal anchor or intermediate target of monetary policy. These frameworks are associated with exchange rate arrangements with no separate legal tender, currency board arrangements, pegs (or stabilized arrangements) with or without bands, crawling pegs (or crawl-like arrangements), and other managed arrangements."
- Monetary aggregate target
  - "The monetary authority uses its instruments to achieve a target growth rate for a monetary aggregate, such as reserve money, M1, or M2, and the targeted aggregate becomes the nominal anchor or intermediate target of monetary policy."
- Inflation-targeting framework
  - "This involves the public announcement of numerical targets for inflation, with an institutional commitment by the monetary authority to achieve these targets, typically over a medium-term horizon. Additional key features normally include increased communication with the public and the markets about the plans and objectives of monetary policymakers and increased accountability of the central bank for achieving its inflation objectives. Monetary policy decisions are often guided by the deviation of forecasts of future inflation from the announced inflation target, with the inflation forecast acting (implicitly or explicitly) as the intermediate target of monetary policy."
- Other
  - "The country has no explicitly stated nominal anchor, but rather monitors various indicators in conducting monetary policy. This category is also used when no relevant information on the country is available."

Source: IMF (2024)

IMF Monetary Policy Frameworks Classification

Table 5. Monetary Policy Frameworks and Exchange Rate Anchors, 2015–23

(Percent of IMF members as of April 30)[1]

	US Dollar	Euro	Composite	Other Currency	Monetary Aggregate	Inflation-Targeting	Other[2]
2015	22.0	13.1	6.3	4.2	13.1	18.8	22.5
2016[3]	20.3	13.0	4.7	4.7	12.5	19.8	25.0
2017	20.3	13.0	4.7	4.7	12.5	20.8	24.0
2018	19.8	13.0	4.7	4.7	12.5	21.4	24.0
2019	19.8	13.0	4.2	4.7	13.5	21.4	23.4
2020	19.8	13.0	4.2	4.7	11.5	22.4	24.5
2021[4]	19.2	13.5	4.1	4.7	13.0	23.3	22.3
2022[5]	19.1	13.4	4.1	5.2	12.9	23.2	22.2
2023	19.6	12.9	4.1	5.2	13.4	23.2	21.6

About 40% of all countries in the world target their FX

Source: IMF (2024)

Examples of countries with fixed FX arrangements

Exchange Rate Arrangement (Number of Countries)	Monetary Policy Framework								
	Exchange Rate Anchor						Monetary Aggregate Target (26)	Inflation-Targeting Framework (45)	Other[1] (42)
	US Dollar (38)		Euro (25)		Composite (8)	Other (10)			
No separate legal tender (14)	Ecuador El Salvador Marshall Islands Micronesia	Palau Panama Timor-Leste	Andorra Kosovo San Marino Montenegro			Kiribati Nauru Tuvalu			
Currency board (12)	Djibouti Hong Kong SAR **ECCU** Antigua and Barbuda Dominica Grenada	St. Kitts and Nevis St. Lucia St. Vincent and the Grenadines	Bosnia and Herzegovina Bulgaria			Brunei Darussalam Macao SAR			
Conventional peg (40)	Aruba The Bahamas Bahrain Barbados Belize Curaçao and Sint Maarten Eritrea	Iraq Jordan Oman Qatar Saudi Arabia Turkmenistan United Arab Emirates	Cabo Verde Comoros Denmark[2] São Tomé and Príncipe **WAEMU** Benin Burkina Faso Côte d'Ivoire Guinea-Bissau Mali Niger Senegal Togo	**CEMAC** Cameroon Central African Rep. Chad Rep. of Congo Equatorial Guinea Gabon	Fiji Libya	Bhutan Eswatini Lesotho Namibia Nepal	Samoa[3]		
Stabilized arrangement (22)	Guyana Honduras (2/22) Lebanon	Maldives Trinidad and Tobago Ukraine 2/22	North Macedonia		Vietnam[4] (10/22)		Bolivia[4] Guinea[4] (9/22) Myanmar[4,8] (11/21) Papua New Guinea[4] Tajikistan[4,6,9] (7/22) Tanzania[4]	Armenia[5] (10/22) Guatemala[4] Romania[5,8] (9/21) Serbia[5]	Azerbaijan[4] Malawi[4] Mozambique[4,6] Sudan[4]

Source: IMF (2024)